

# **TITLE PAGE**

- **Problem Statement ID – 1**
- **Problem Statement Title - Eco-Loop Building Agents: Autonomous AI-Driven Energy Optimization for Smart Buildings**
- **Theme- Smart Buildings / Sustainability / Physical AI**
- **PS Category- Software**
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# ECO-LOOP BUILDING AGENTS

## Proposed Solution

- A closed-loop Physical AI system where an open-source LLM (Qwen2.5) reads live sensor data from a running EnergyPlus building simulation and autonomously adjusts HVAC setpoints in real time via the EMS API
- Turns a passive, rule-based building into a self-correcting agent that continuously balances energy efficiency with occupant thermal comfort
- Traditional Building Management Systems use rigid, fixed schedules that can't adapt to real-time weather, occupancy, or grid conditions, wasting energy
- Our agent ingests live zone temperature, energy use, and thermal comfort (PMV) data every cycle and reasons about targets before pushing new setpoints straight back into the live simulation
- Unlike hardcoded control scripts, our LLM genuinely decides which control tool to call and with what value, using real function/tool-calling, not scripted if-else logic

# TECHNICAL APPROACH

- EnergyPlus (DOE physics-based simulation engine) + EMS Python API (pyenergyplus) for live, bidirectional sensor/actuator access during a running simulation
- Open-source LLM (Qwen2.5-7B-Instruct) served locally via Ollama, using genuine function/tool-calling as the reasoning brain
- Python unified codebase: EnergyPlus bridge, custom agentic tool layer(get\_zone\_status, set\_cooling\_setpoint, set\_heating\_setpoint, get\_recent\_log\_errors), and orchestrator
- Methodology: sensor read → state sent to LLM → LLM reasons and calls a tool → setpoint injected into the live sim → logged

## FEASIBILITY AND VIABILITY

- Fully open-source stack (EnergyPlus, Ollama, Qwen2.5) runs entirely on local hardware — no paid APIs, no cloud dependency, zero recurring cost
- Challenges: LLM tool-calling reliability, control loop latency vs. simulation speed, and ensuring the system never crashes over long, multi-day runs
- Strategies for overcoming these challenges: batching LLM calls across multiple timesteps to manage latency, enforcing safe setpoint bounds in every control tool, and wrapping all EMS/LLM calls in retry and error-handling logic for resilience

# ARTIFACTS

- Relevant artifacts, such as :

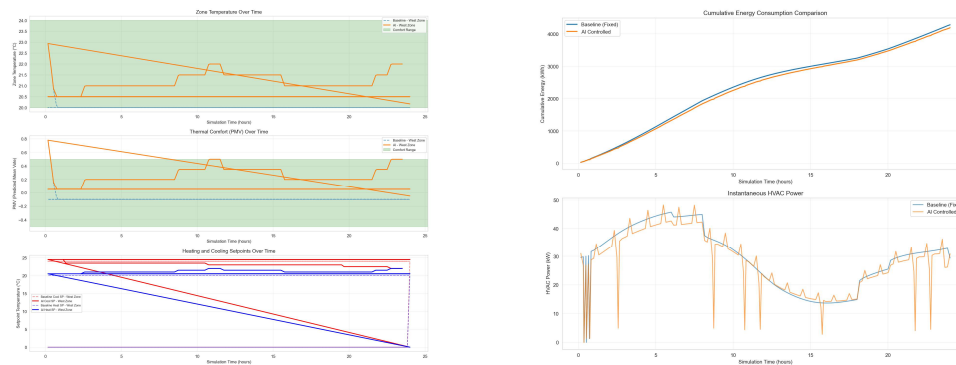
- GitHub repo: <https://github.com/PrachiPatel2105/eco-loop-building-agents>
- live log entries showing LLM reasoning text + tool call (e.g. set\_cooling\_setpoint) alongside the EnergyPlus run

```
LLM Decision Point - 1986-01-01 08:30
=====
West Zone: 21.8°C, PMV=0.15, HVAC=4.20kW

LLM Reasoning:
"Zone comfortable but HVAC power is 4.2 kW. Since comfort
is satisfied, widening deadband to reduce energy..."

Executing 2 tool call(s):
1. set_cooling_setpoint({"zone": "West Zone", "value": 25.0})
2. set_heating_setpoint({"zone": "West Zone", "value": 19.5})
```

- baseline vs AI-controlled kWh and comfort compliance charts from analyze\_results.py



# RESEARCH AND REFERENCES

- EnergyPlus & EMS Application Guide — U.S. Department of Energy ([energyplus.net](http://energyplus.net))
- Ollama documentation & Qwen2.5 model card ([ollama.com](http://ollama.com), Alibaba Qwen team)
- ANSI/ASHRAE Standard 55 & ISO 7730 — thermal comfort and PMV/PPD methodology
- U.S. DOE — Buildings consume ~40% of global energy and CO<sub>2</sub> emissions